

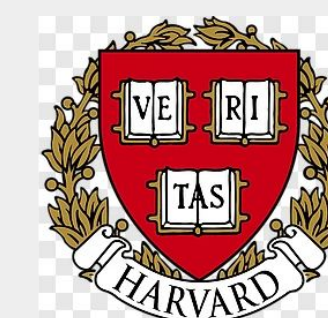
Large Language Model Fingerprints From Normal Interaction



Annesya Banerjee^{1,2}, Itay Lavie³

annesya_banerjee@g.harvard.edu; itaylavie@g.harvard.edu

¹PhD Program in SHBT, Harvard; ²Dept. of Brain and Cognitive Science, MIT; ³School of Engineering and Applied Sciences, Harvard



Introduction

The Challenge:

- Users cannot verify which LLM they're interacting with in third-party applications
- LLMs pose risks: bias amplification, hallucinations, privacy concerns
- Need for attribution and accountability

Our Solution:

A classifier that identifies LLMs from their text output embeddings with **89% accuracy**

Key Requirements:

- Intra-model consistency (same classification across versions)
- Inter-model discrepancy (distinguish similar models)
- Generalization to unseen versions and semantic domains

Data Collection:

We created a dataset of 4410 general questions and collected responses from 7 different commercial LLMs.

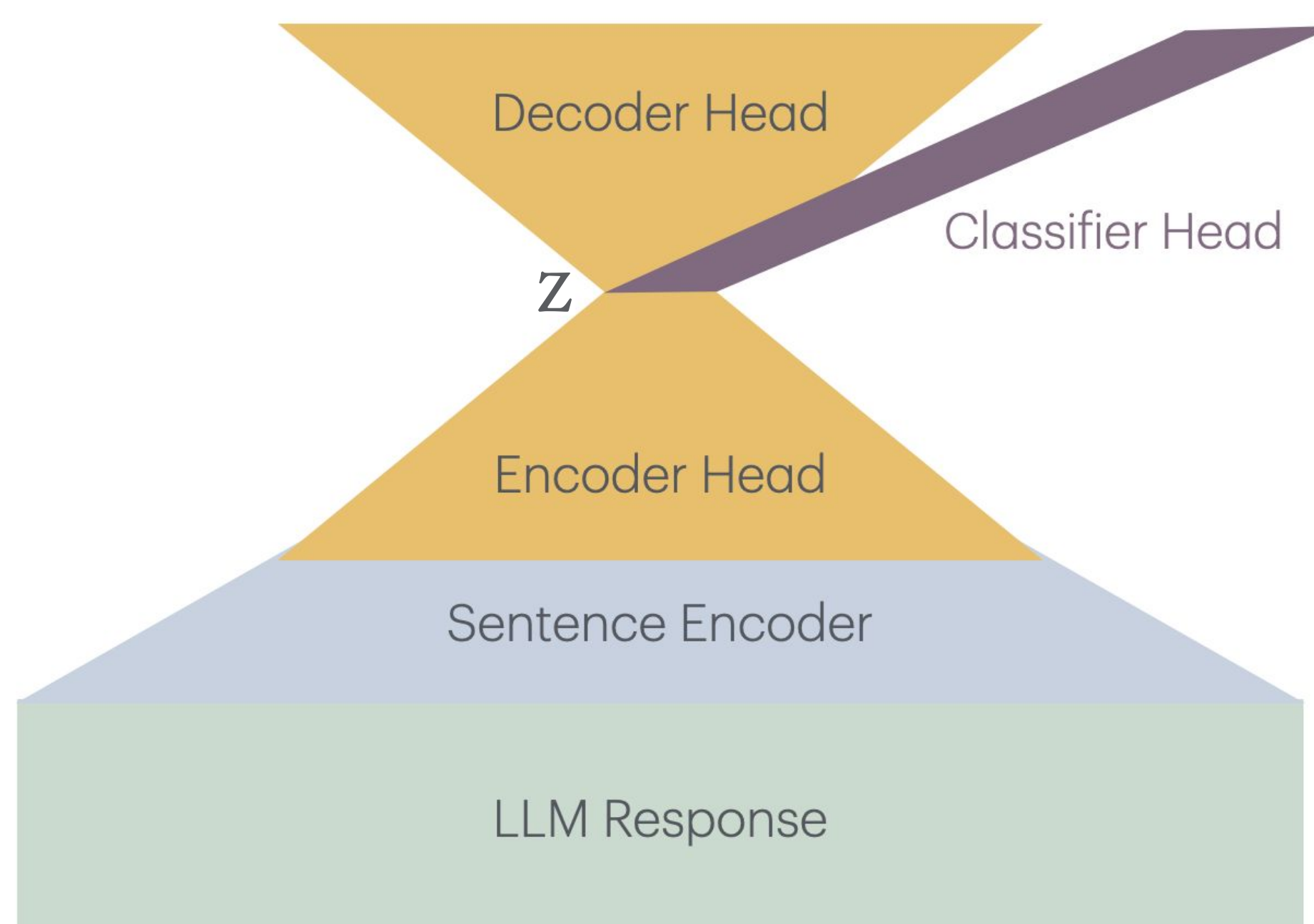
Pre-Processing:

We embedded each response to a dense vector using a MiniLM - a sentence encoder

Classifier:

- Encoder - create a low dimensional discriminative representation of the data
- Decoder - reconstruction task to avoid information loss and representation collapse
- Classifier - classification task to create a discriminative representation

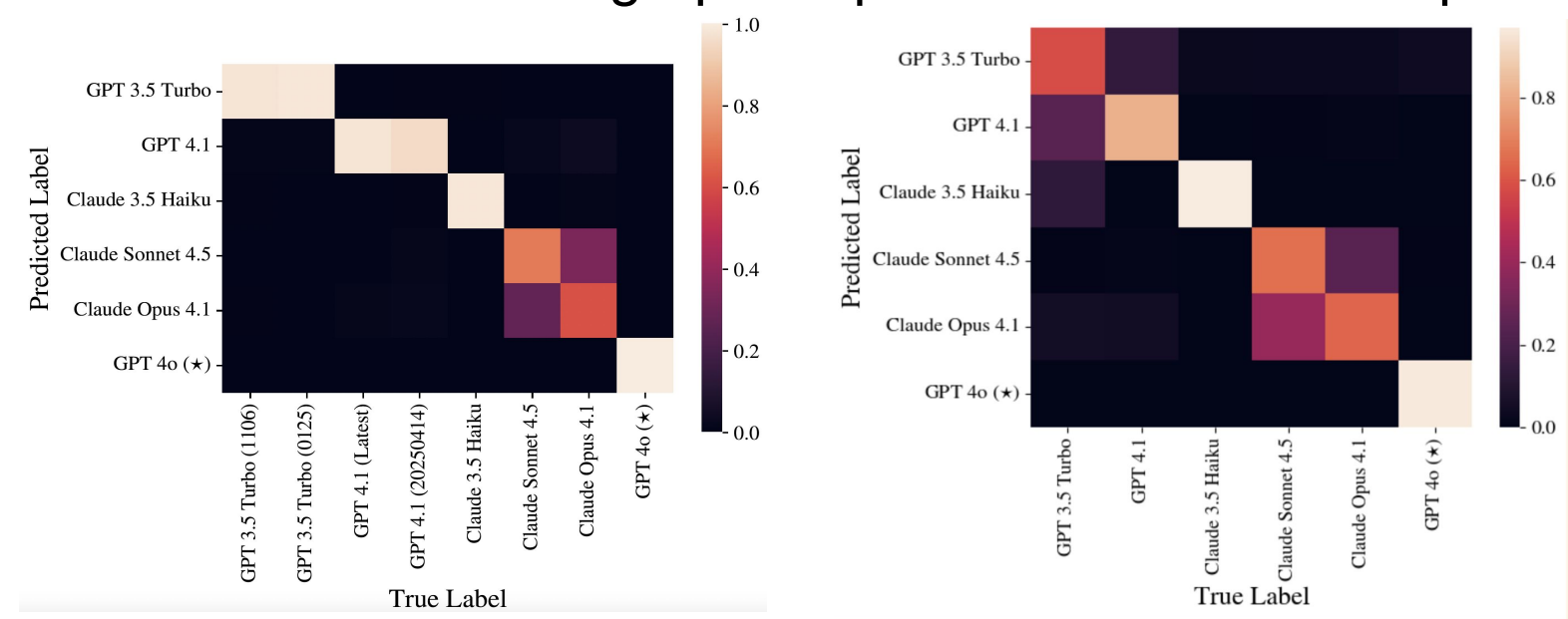
Methods



Experimental Results

Inter- and Intra-model discrimination

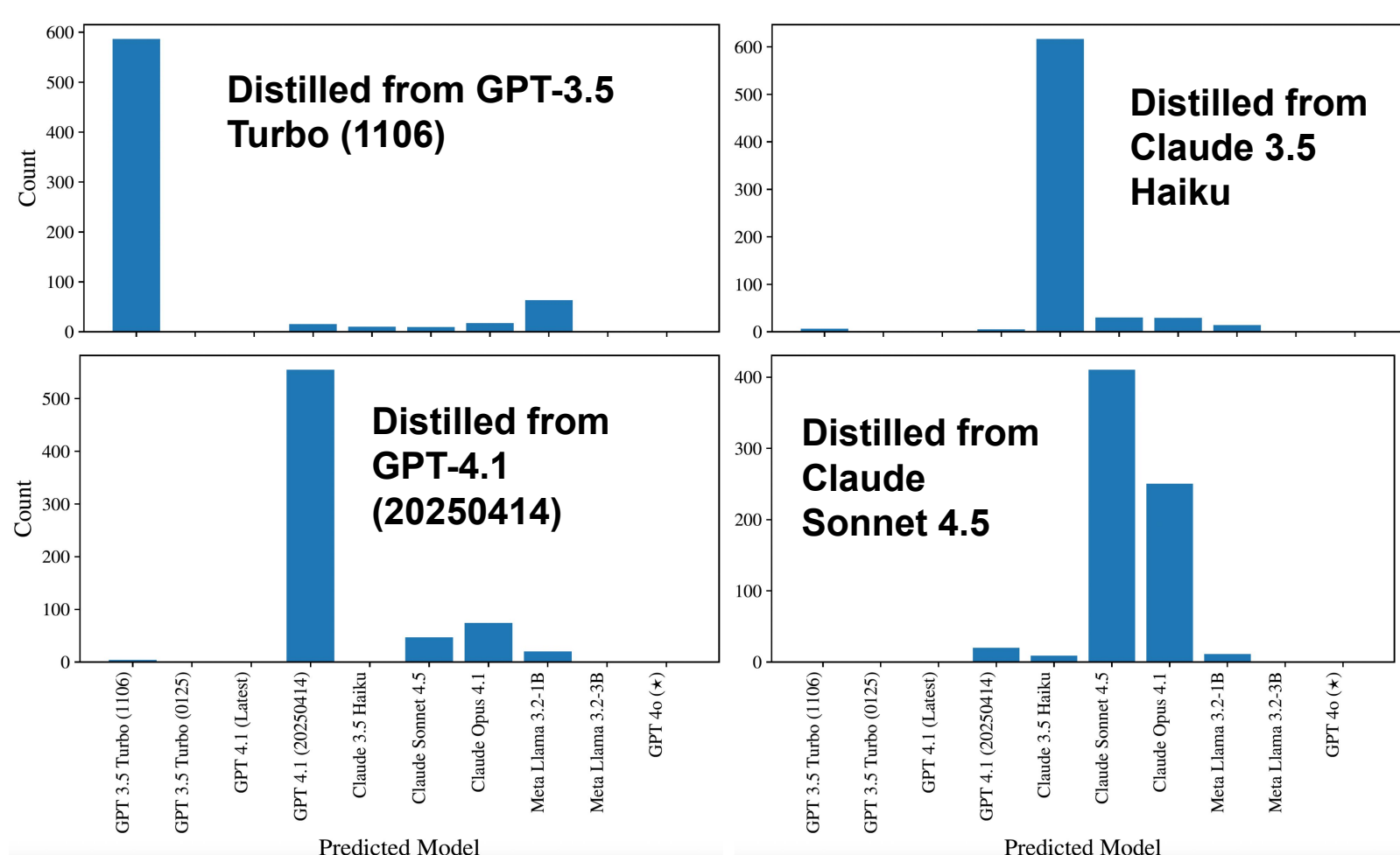
- **89% overall accuracy** on test set
- Successfully generalizes to unseen versions (GPT-3.5 0125, GPT-4.1 Latest)
- Version-invariant fingerprints persist across model updates



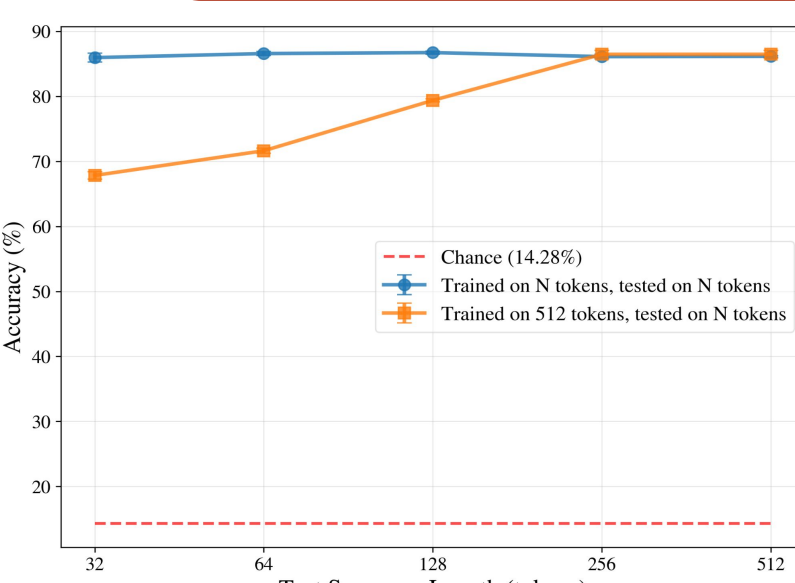
Confusion matrices. Left: Results when also tested on held-out model classes (GPT-3.5 Turbo (0125), GPT-4.1 (Latest)). **Right:** k-means classification when tested on held-out model classes without seeing any version of GPT-3.5 Turbo during training

Distilled fingerprint dominated by teacher model

- Llama-3.2-1B fine-tuned on GPT/Claude outputs using LoRA
- Distilled models classified as **teacher, not base architecture**
- Behavioral fingerprints transfer through distillation
- Further work needed to improve fingerprint method and avoid model usage abuse
- Implications for model provenance and IP protection

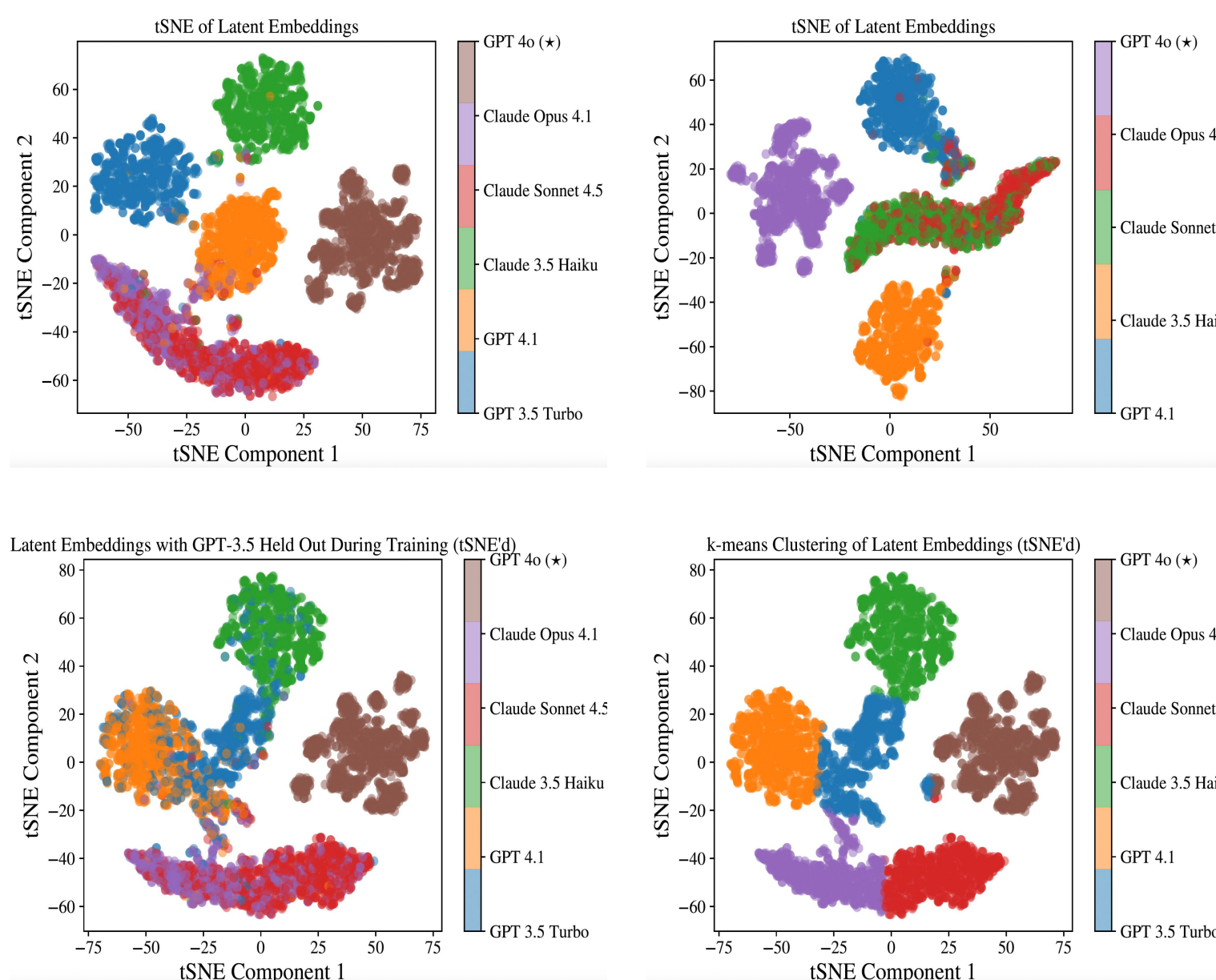


Robust performance across sequence length



- **86-87% accuracy** even with 32-token responses
- Single classifier trained on 512 tokens generalizes to shorter sequences
- Fingerprints manifest early in generation

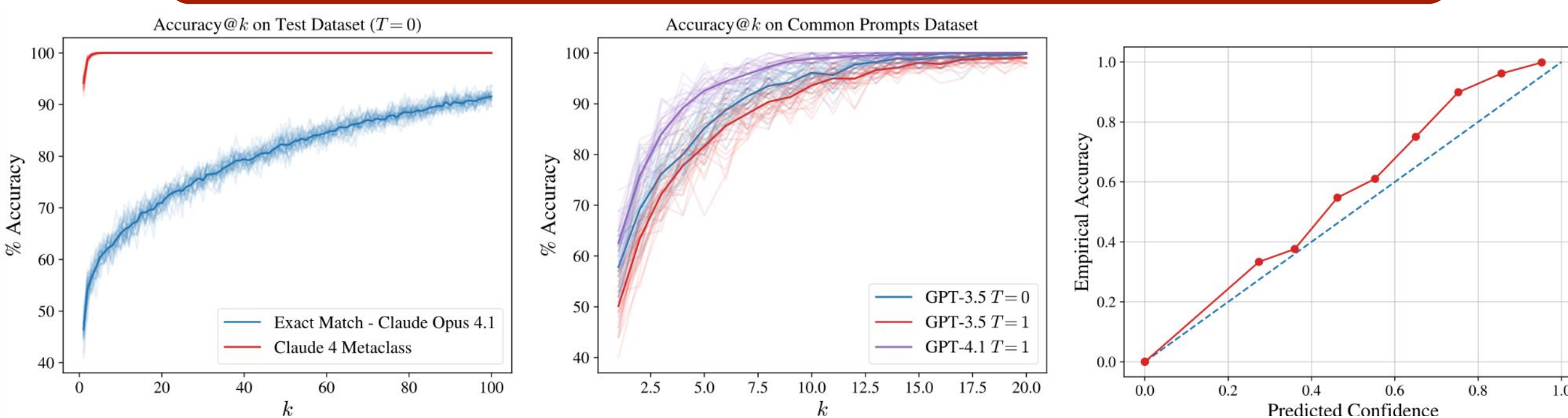
Model-specific low-dimensional cluster formation



- t-SNE visualization shows well-separated clusters
- k-means successfully identifies OOD models (e.g., GPT-3.5 held out during training)
- Geometric structure enables interpretable predictions
- Specifically important for OOD model fingerprinting

2-dimensional t-SNE of the encoded discriminatory representation. Top left: training the model on all training models in the dataset; Top right: leaving out GPT-3.5 during training and inference; Bottom left: leaving out GPT-3.5 during training and including it during inference (ground truth classes); Bottom right: k-means classification while leaving out GPT-3.5 during training and including it during inference.

Test-time scaling improves performance



We use k samples from the same LLM to achieve better classification

- Improves classification for “hard” cases where single sample classification accuracy is low
- Enables classification in real-world situations: out-of-distribution prompt-response from a dataset of common prompts and sampling with temperature
- Theory motivated algorithms for $k>1$ (multiple responses)

Conclusion

Key Contributions:

- Dataset of 4,410 prompts across 7 commercial LLMs
- 89% accuracy with robust generalization to unseen versions; further improvement with test-time scaling enabled with multiple samples; robust performance across sequence lengths
- Distillation transfers behavioral fingerprints from teacher models; might require more work to avoid model usage abuse

Future Directions:

- Adversarial robustness evaluation; Multilingual and domain-specific performance; Continuous learning for new model releases

Code available at <https://github.com/ItayXD/LLM-fingerprints.git>

Key References: Fingerprinting: Pasquini+ '25, Fu+ '25, Nikolic+ '25 | Ownership: Nasery+ '25, Gubri+ '24, Xu+ '24 | Applications: Busch+ '25, Yu+ '24 | Methods: Wang+ '20 (MiniLM), Wen+ '16 (Center Loss), Hu+ '21 (LoRA)